Neck vessels & nerves

Carotid and jugular systems, the subclavian, the last four cranial nerves, cervical plexus, phrenic nerve and sympathetic chain.

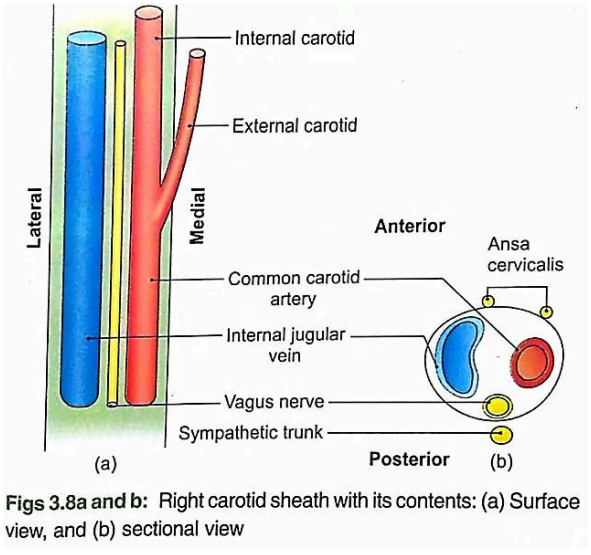
1. Carotid sheath and common carotid
2. External carotid artery
3. Internal carotid artery
4. Veins of the neck
5. Root of the neck: subclavian artery and thoracic duct
6. Glossopharyngeal nerve (IX)
7. Vagus nerve (X)
8. Accessory nerve (XI) and hypoglossal nerve (XII)
9. Cervical plexus, ansa cervicalis and phrenic nerve
10. Cervical sympathetic chain
11. In theatre
12. On imaging

01 Carotid sheath and common carotid

- Sheath contains the **common/internal carotid medially, IJV laterally, vagus between and behind them**; the ansa cervicalis lies on or in its front; the sympathetic chain is behind it.
- **Common carotid:** right from the brachiocephalic trunk, left from the arch. Ascends to the **upper border of the thyroid cartilage (C3–C4)**, where it divides. No branches in the neck.
- **Carotid sinus** (baroreceptor) at the ICA origin and **carotid body** (chemoreceptor) in the bifurcation, both supplied by **IX** (nerve of Hering).

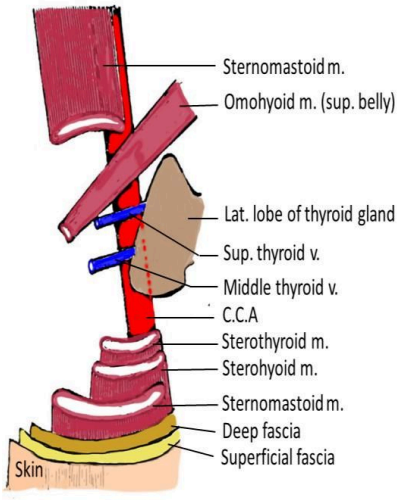
CLINICAL PEARL Carotid sinus massage slows the heart; a carotid body tumour splays the ICA and ECA apart (lyre sign).

SOURCES DIFFER ORDER IN THE SHEATH: Doss: CCA anteromedial, IJV posterolateral, vagus between and behind (as here). ENT Steps lists superficial to deep as IJV, vagus, CCA.

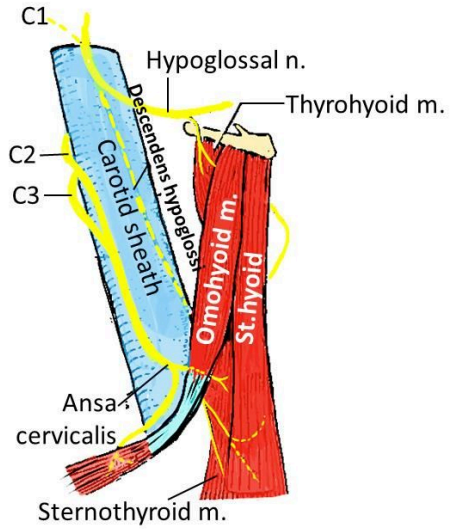


Figs 3.8a and b: Right carotid sheath with its contents: (a) Surface view, and (b) sectional view

Carotid sheath and its contents
ENT Steps Anatomy, p. 266

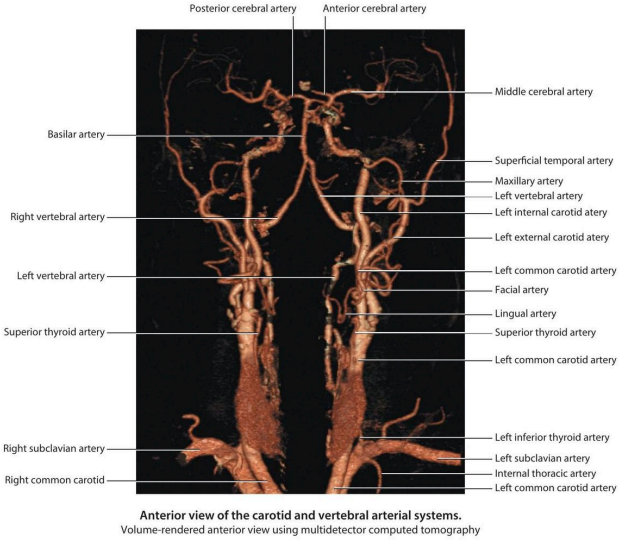


Relations of the common carotid artery
Dr. Sameh Doss handout, p. 136



Ansa cervicalis and the carotid sheath

Dr. Sameh Doss handout, p. 124



Carotid and vertebral arteries: CT angiography, anterior view

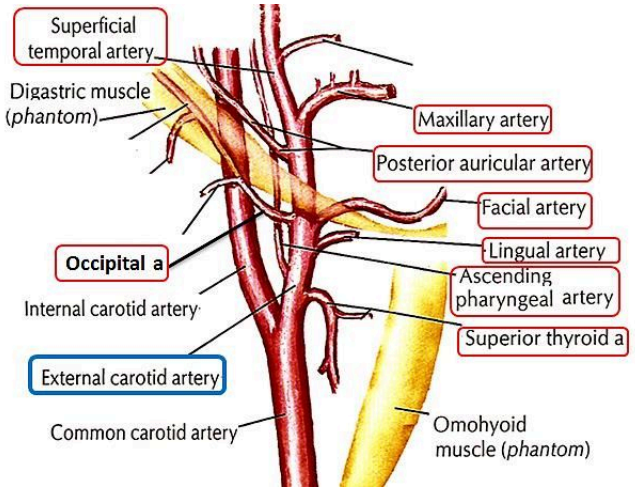
Gray's Atlas of Anatomy, 2nd ed., p. 501

02 External carotid artery

- Begins in front of and medial to the ICA; ends in the parotid behind the neck of the mandible by dividing into the **maxillary** and **superficial temporal** arteries.
- **Branches ("Some Anatomists Like Freaking Out Poor Medical Students"):** Superior thyroid, Ascending pharyngeal, Lingual, Facial, Occipital, Posterior auricular, Maxillary, Superficial temporal.
- The ECA has branches in the neck; the ICA does not. This is how you tell them apart before ligation.
- **Lingual artery:** loops above the greater horn of the hyoid and runs **deep to hyoglossus**, while the hypoglossal nerve runs **on** the hyoglossus.
- **Facial artery:** grooves the submandibular gland, then crosses the mandible at the front of the masseter.

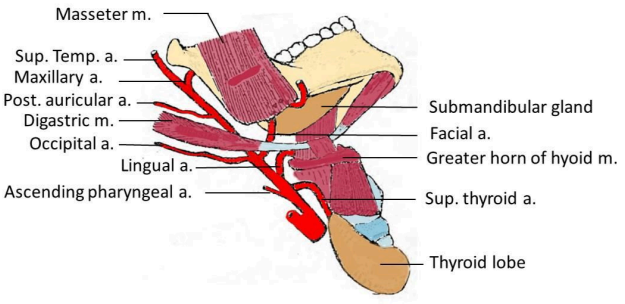
CLINICAL PEARL Hypoglossal nerve crosses superficial to both carotids, looping round the occipital artery's SCM branch: protect it when ligating the ECA.

SOURCES DIFFER FIRST BRANCH: ENT Steps and Gray's say the superior thyroid; Doss calls the ascending pharyngeal the first branch (from the medial side).



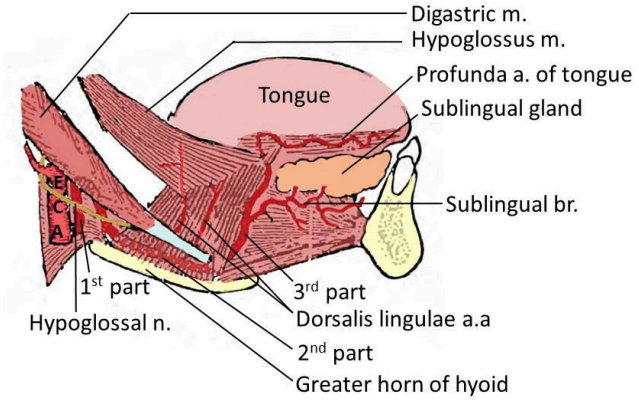
Branches of the external carotid artery

ENT Steps Anatomy, p. 267



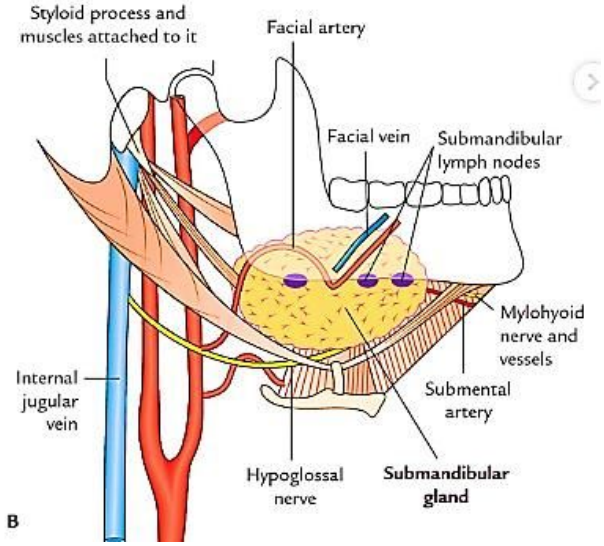
Branches of the external carotid artery

Dr. Sameh Doss handout, p. 142



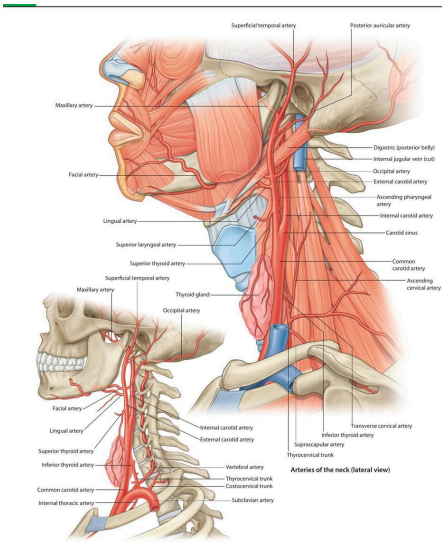
Lingual artery and hypoglossal nerve

Dr. Sameh Doss handout, p. 142



Submandibular triangle: facial vessels and hypoglossal nerve

ENT Steps Anatomy, p. 271



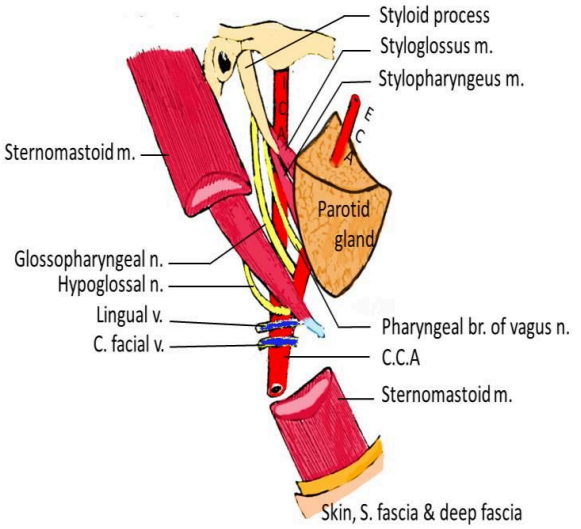
Arteries of the neck (lateral view)

Gray's Atlas of Anatomy, 2nd ed., p. 546

03 Internal carotid artery

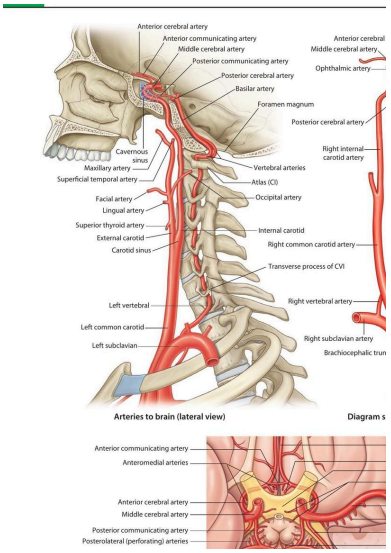
- Cervical part runs up in the carotid sheath to the carotid canal, **without branches**. Then petrous, lacerum, cavernous and cerebral parts.
- **Behind it:** longus capitis, superior cervical ganglion, superior laryngeal nerve. **Between ICA and ECA:** styloglossus, stylopharyngeus, IX and the pharyngeal branch of X.
- Lies about 2.5 cm behind and lateral to the tonsil; aberrant medial loops can lie under the posterior pharyngeal wall.

> **CLINICAL PEARL** A pulsatile tonsil or posterior pharyngeal wall can be an aberrant ICA: do not incise it.



Internal carotid artery in the neck

Dr. Sameh Doss handout, p. 138

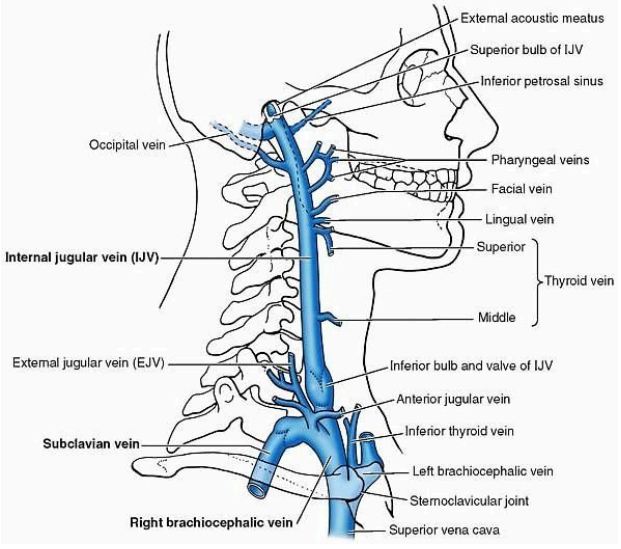


Arteries to the brain (lateral view): carotid and vertebral systems

Gray's Atlas of Anatomy, 2nd ed., p. 500

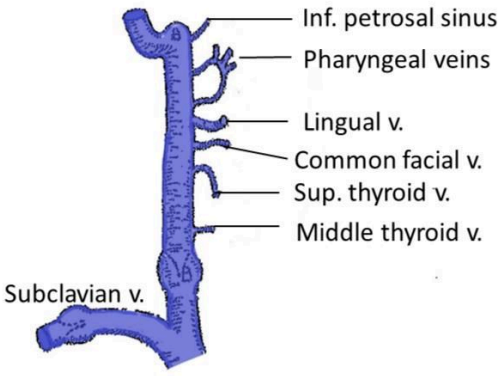
04 Veins of the neck

- **IJV:** continuation of the **sigmoid sinus** at the jugular foramen; runs in the carotid sheath to join the subclavian vein behind the sternoclavicular joint (forming the brachiocephalic vein).
- **Tributaries:** inferior petrosal sinus, pharyngeal, common facial, lingual, superior and middle thyroid veins.
- **External jugular vein:** from the posterior division of the retromandibular vein and the posterior auricular vein; crosses SCM to drain into the subclavian vein.
- **Anterior jugular veins** join across the suprasternal space (jugular arch).



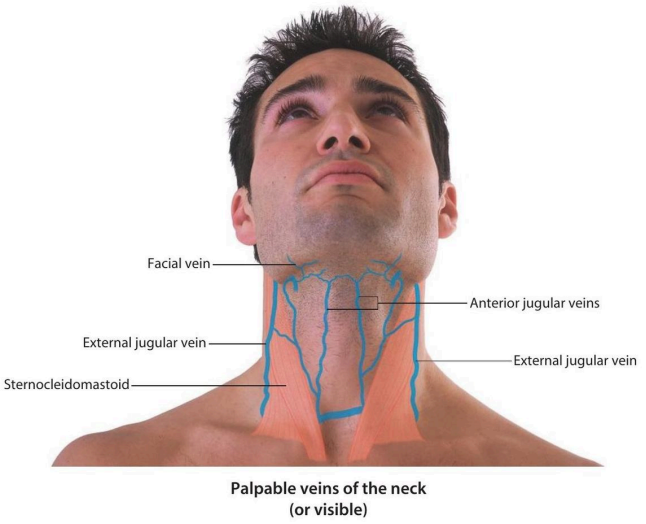
Veins of the head and neck

ENT Steps Anatomy, p. 268



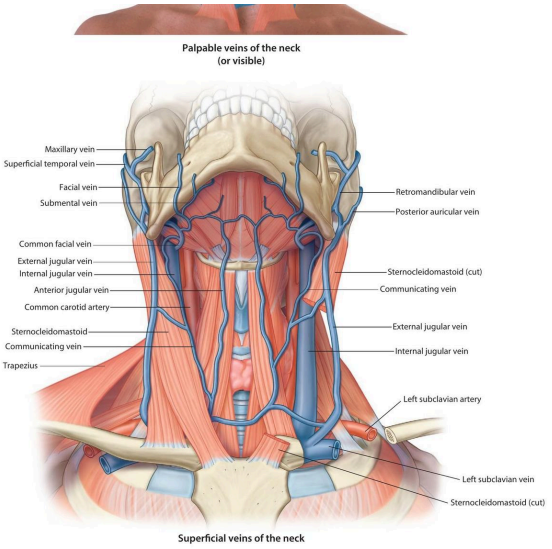
Tributaries of the internal jugular vein

Dr. Sameh Doss handout, p. 146



Palpable veins of the neck

Gray's Atlas of Anatomy, 2nd ed., p. 539



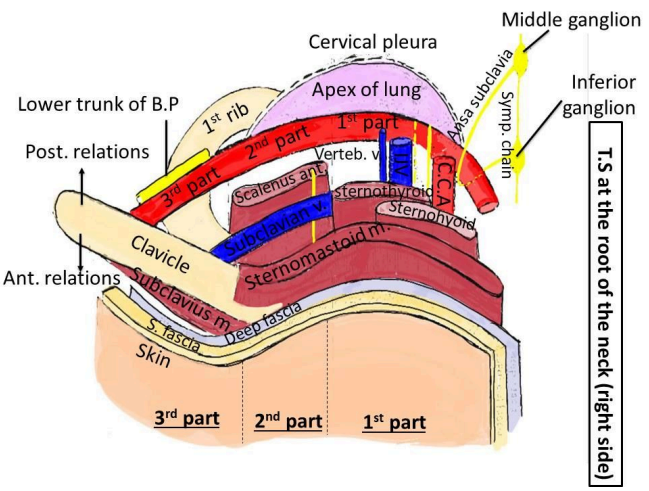
Superficial veins of the neck

Gray's Atlas of Anatomy, 2nd ed., p. 539

05 Root of the neck: subclavian artery and thoracic duct

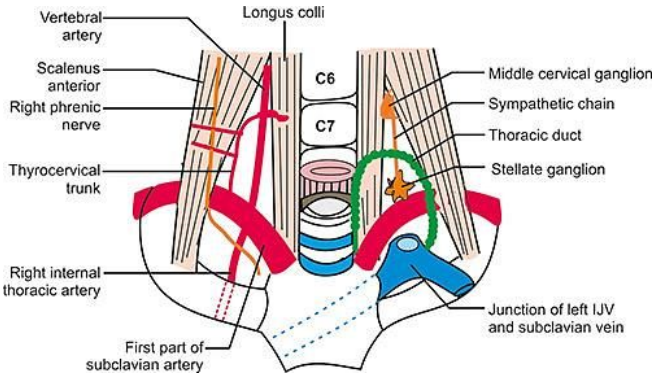
- **Subclavian artery:** first part (medial to scalenus anterior) gives the **vertebral**, **internal thoracic** and **thyrocervical trunk** (inferior thyroid, transverse cervical, suprascapular); second part (behind the muscle) gives the costocervical trunk; third part (lateral) usually none.
- **Inferior thyroid artery** arches medially behind the carotid sheath to the back of the thyroid, crossing the RLN.
- **Thoracic duct** arches on the left at C7 to enter the junction of the IJV and subclavian vein.
- **Chassaignac's (scalenovertebral) triangle:** longus colli, scalenus anterior and subclavian artery, apex at the C6 tubercle.

CLINICAL PEARL Milky drainage after a left neck dissection means a chyle leak from the thoracic duct.



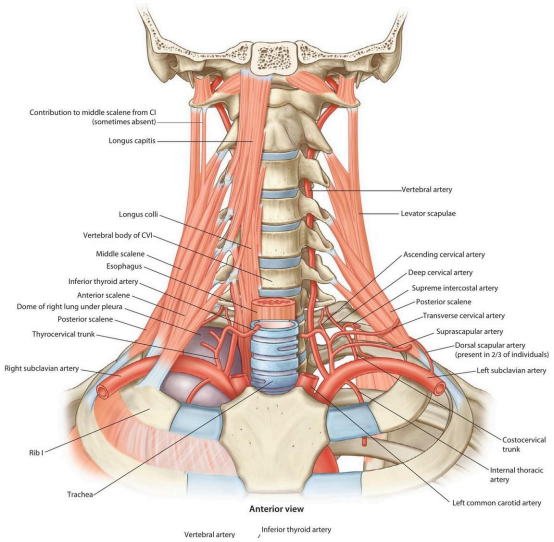
Subclavian artery and its three parts

Dr. Sameh Doss handout, p. 131



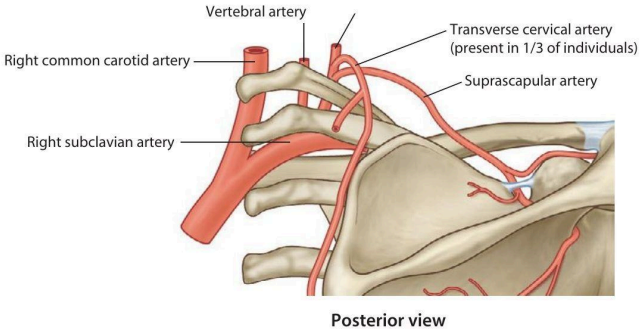
Root of the neck: subclavian artery, thoracic duct, phrenic nerve

ENT Steps Anatomy, p. 269



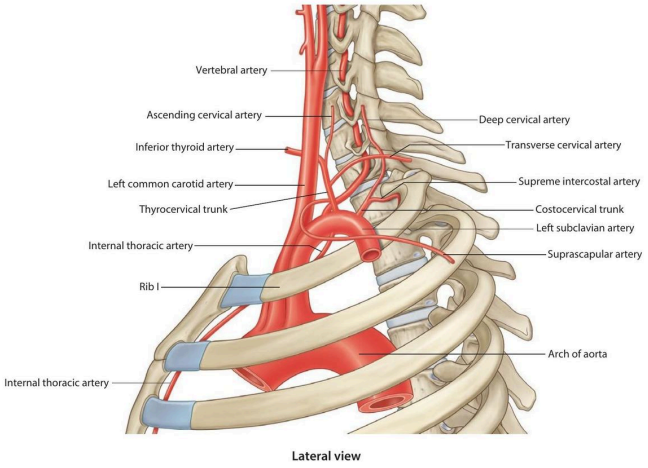
Root of the neck: arteries (anterior view)

Gray's Atlas of Anatomy, 2nd ed., p. 548



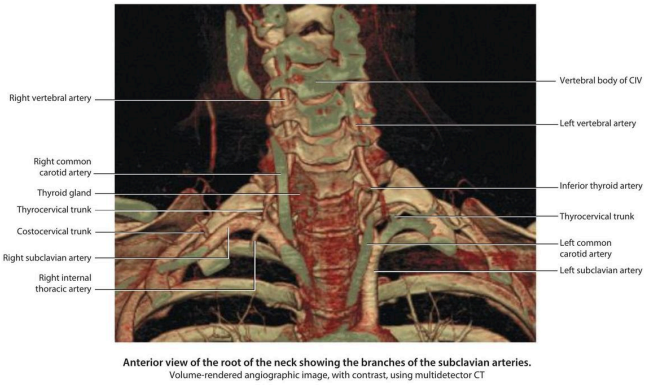
Root of the neck: thyrocervical trunk (posterior view)

Gray's Atlas of Anatomy, 2nd ed., p. 548



Root of the neck: arteries (lateral view)

Gray's Atlas of Anatomy, 2nd ed., p. 549



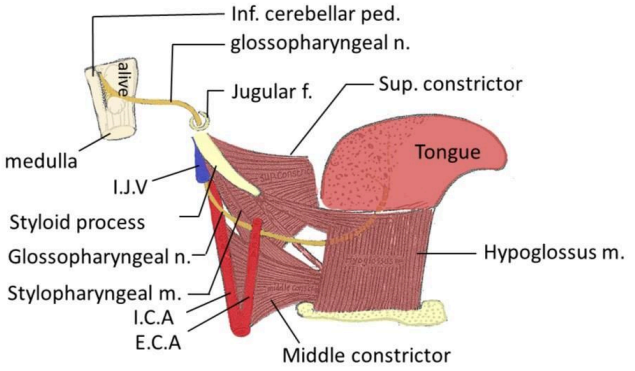
Branches of the subclavian arteries: CT angiography

Gray's Atlas of Anatomy, 2nd ed., p. 549

06 Glossopharyngeal nerve (IX)

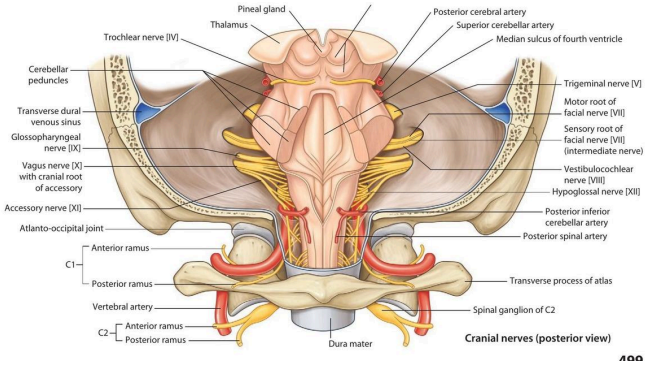
- Leaves through the jugular foramen; curves forward between the ICA and ECA on **stylopharyngeus** (its only muscle) to the tongue base.
- **Branches:** tympanic (Jacobson's) → lesser petrosal → parotid; carotid sinus nerve; pharyngeal (sensory to the plexus); tonsillar; lingual (taste and sensation, posterior third).
- Afferent limb of the **gag reflex**.

> **CLINICAL PEARL** Glossopharyngeal neuralgia: lancinating pain in the tonsil and ear on swallowing.



Glossopharyngeal nerve

Dr. Sameh Doss handout, p. 151



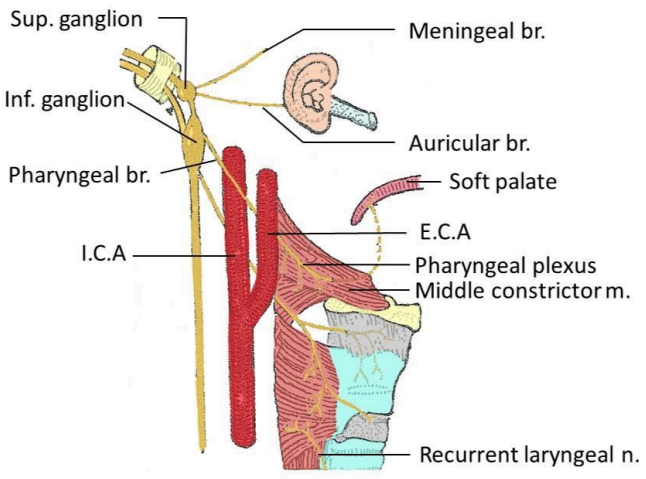
Cranial nerves (posterior view): IX, X, XI and XII at the jugular foramen

Gray's Atlas of Anatomy, 2nd ed., p. 499

07 Vagus nerve (X)

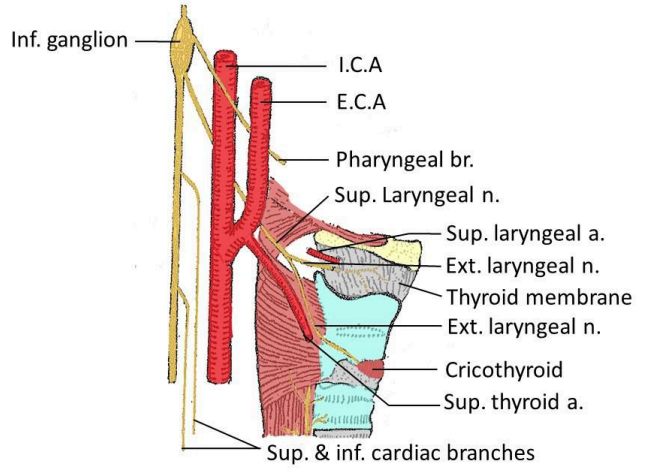
- Leaves the jugular foramen with **superior and inferior ganglia**; descends in the carotid sheath behind and between the artery and vein.
- **Branches in the neck:** auricular (Arnold's), pharyngeal (motor to the pharyngeal plexus, soft palate except tensor), **superior laryngeal** (internal and external), cardiac branches, **recurrent laryngeal**.
- Right RLN hooks under the subclavian artery; left under the arch of the aorta.

CLINICAL PEARL A high vagal lesion (above the pharyngeal branch) also paralyses the palate: the uvula deviates to the healthy side.



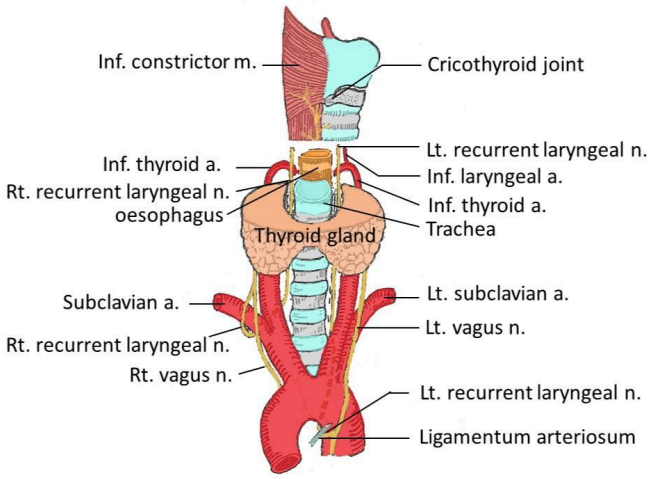
Vagus nerve in the neck

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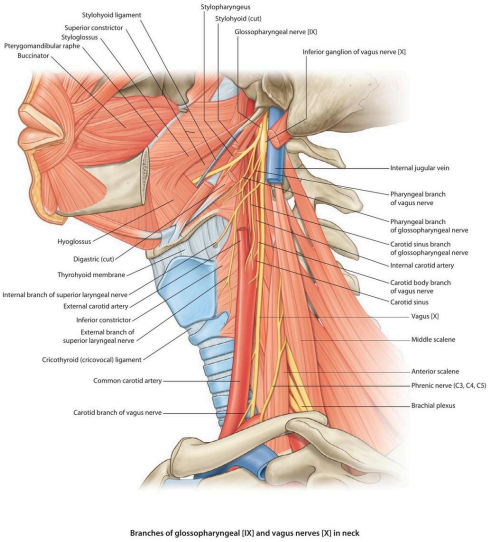
Branches of the vagus nerve in the neck

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Recurrent laryngeal nerves and inferior thyroid arteries

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Branches of glossopharyngeal [IX] and vagus nerves [X] in neck

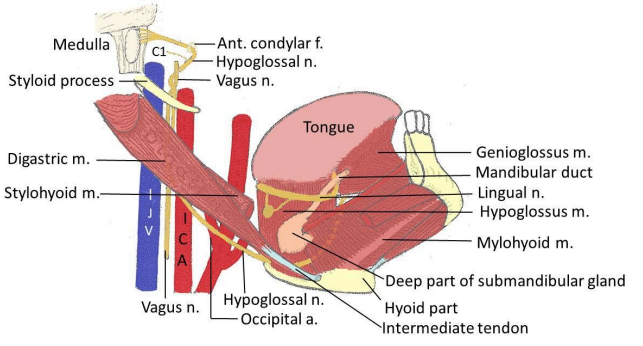
Branches of the glossopharyngeal [IX] and vagus [X] nerves in the neck

Gray's Atlas of Anatomy, 2nd ed., p. 543

08 Accessory nerve (XI) and hypoglossal nerve (XII)

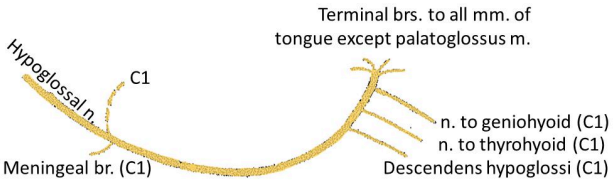
- **XI (spinal part):** from C1–C5 up through the foramen magnum and out of the jugular foramen; runs back across (usually in front of) the IJV, deep to the posterior belly of digastric, into **SCM**, then across the posterior triangle to **trapezius**.
- **XII:** leaves the hypoglossal canal, passes between ICA and IJV, loops round the occipital artery, crosses **superficial to the ICA and ECA** above the bifurcation, runs on hyoglossus below the lingual nerve and Wharton's duct.
- C1 fibres travel with XII and leave as the **superior root of the ansa cervicalis**, the nerve to thyrohyoid and geniohyoid.

> **CLINICAL PEARL** In level IIA dissection the accessory nerve is found where it crosses the IJV, just below the posterior belly of digastric.



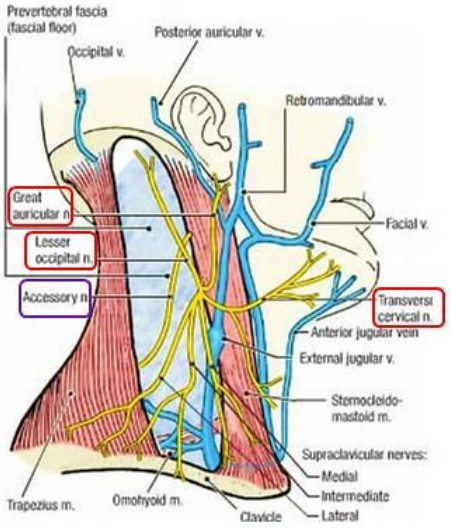
Hypoglossal nerve

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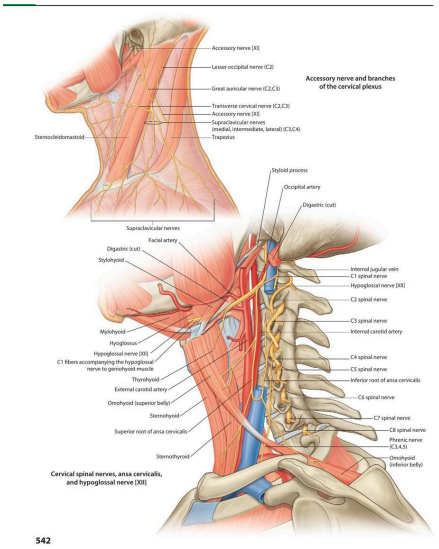
Branches of the hypoglossal nerve

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Nerves of the neck: cervical plexus and accessory nerve

ENT Steps Anatomy, p. 265



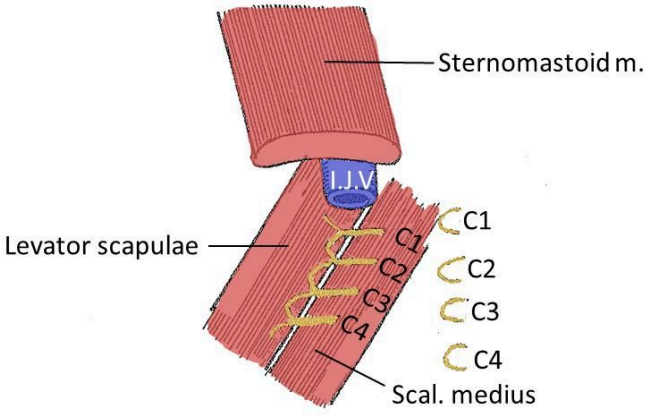
Accessory nerve, cervical plexus, ansa cervicalis and hypoglossal nerve

Gray's Atlas of Anatomy, 2nd ed., p. 542

09 Cervical plexus, ansa cervicalis and phrenic nerve

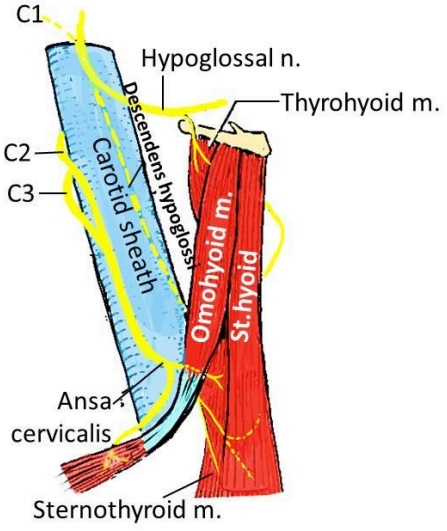
- **Cervical plexus (C1–C4):** cutaneous branches at Erb's point: **lesser occipital (C2), great auricular (C2–3), transverse cervical (C2–3), supraclavicular (C3–4)**.
- **Ansa cervicalis:** superior root (C1 via XII) + inferior root (C2–3); supplies the strap muscles except thyrohyoid.
- **Phrenic nerve (C3–C5):** runs down on **scalenus anterior** deep to the prevertebral fascia, behind the transverse cervical and suprascapular arteries.

> **CLINICAL PEARL** "C3, 4, 5 keeps the diaphragm alive." Staying above the prevertebral fascia protects the phrenic nerve and brachial plexus in neck dissection.



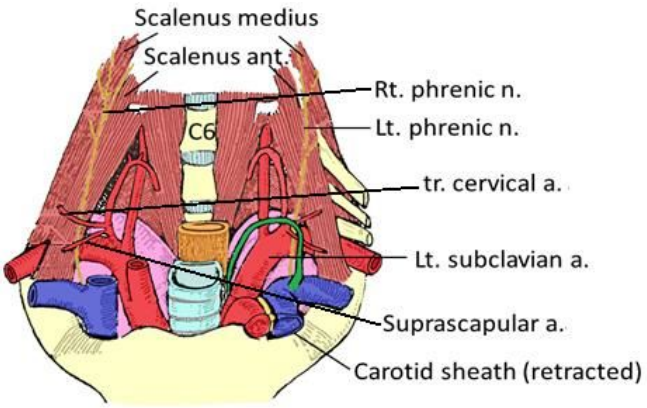
Cervical plexus

Dr. Sameh Doss handout, p. 158



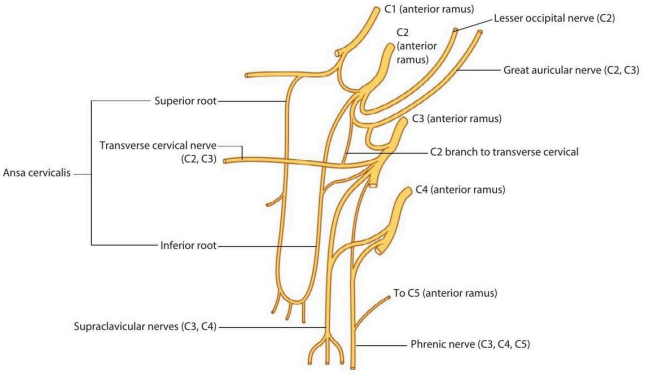
Ansa cervicalis and the carotid sheath

Dr. Sameh Doss handout, p. 124



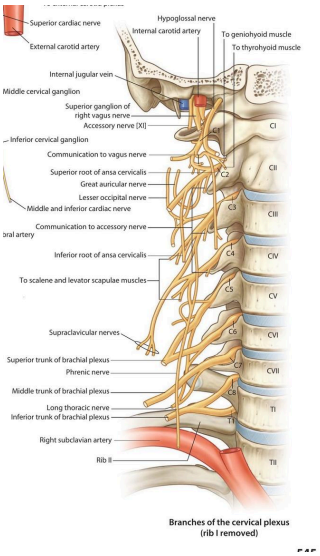
Phrenic nerves on scalenus anterior

Dr. Sameh Doss handout, p. 159



Cervical plexus

Gray's Atlas of Anatomy, 2nd ed., p. 544



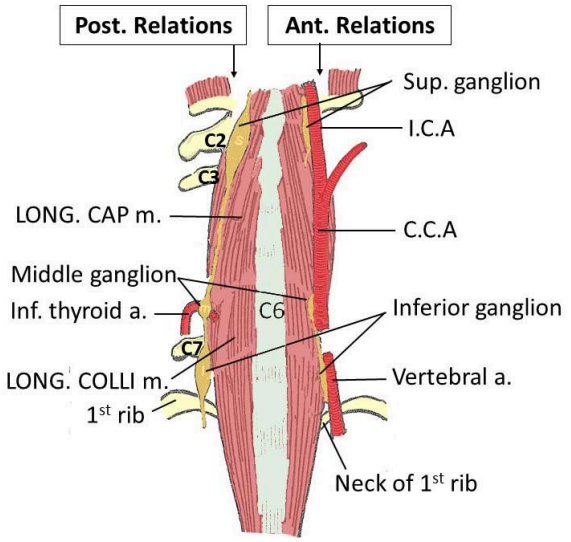
Branches of the cervical plexus (rib I removed)

Gray's Atlas of Anatomy, 2nd ed., p. 545

10 Cervical sympathetic chain

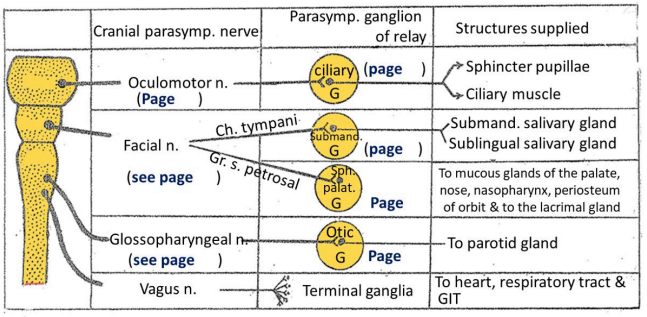
- Lies behind the carotid sheath on longus colli/capitis. **Superior cervical ganglion** (C2–3, largest), **middle** (C6), **inferior** (often fused with T1 as the **stellate ganglion**).
- Postganglionic fibres follow the ICA and ECA to the head: pupil dilator, tarsal muscle, sweat glands, blood vessels, salivary glands.
- Cranial parasympathetic ganglia: ciliary (III), **pterygopalatine (VII)**, **submandibular (VII)**, **otic (IX)**.

> **CLINICAL PEARL** Horner's syndrome (ptosis, miosis, anhidrosis, enophthalmos) can follow neck surgery or a mass compressing the chain.



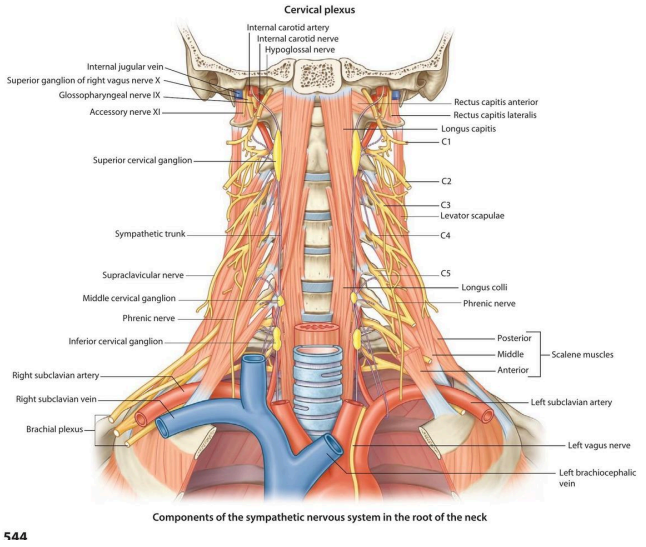
Cervical sympathetic chain

Dr. Sameh Doss handout, p. 160



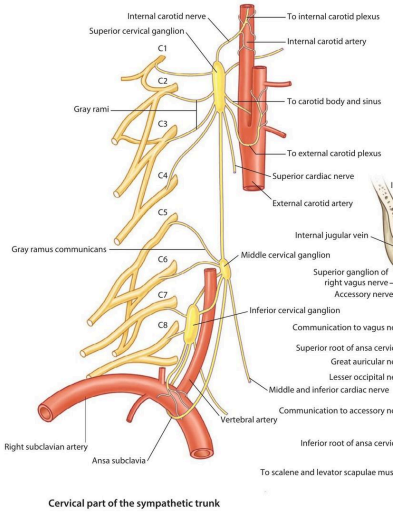
Cranial parasympathetic ganglia

Dr. Sameh Doss handout, p. 159



Sympathetic system in the root of the neck

Gray's Atlas of Anatomy, 2nd ed., p. 544



Cervical part of the sympathetic trunk

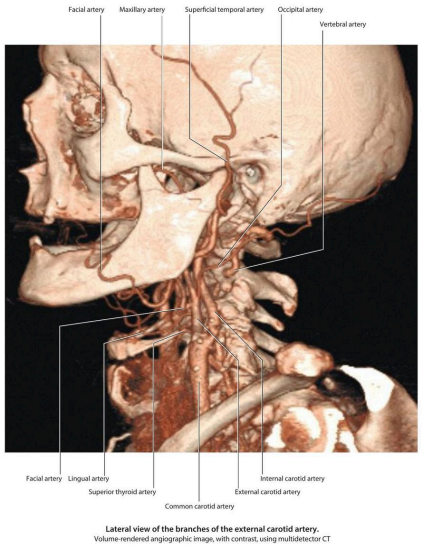
Gray's Atlas of Anatomy, 2nd ed., p. 545

11 Surgical anatomy: In theatre

- **ECA ligation:** confirm branches before tying; tie above the superior thyroid artery.
- **Carotid body tumour:** Shamblin I (small, localised, minimal vessel attachment), II (partly surrounds the vessels), III (encases them); risk to X and XII rises with grade.
- **IJV ligation:** one side is safe; bilateral simultaneous ligation risks facial and cerebral oedema, so stage it.
- **Accessory nerve** at level II: usually lateral (superficial) to the IJV, occasionally medial or through it.

12 Radiological anatomy: On imaging

- **CT/MR angiography:** carotid body tumours splay the ICA and ECA (lyre sign); glomus vagale pushes both forward without splaying.
- **Doppler ultrasound** for carotid disease and IJV thrombosis (Lemierre's syndrome after pharyngitis).



Branches of the external carotid artery: CT angiography

Gray's Atlas of Anatomy, 2nd ed., p. 547

Tables

Branches of the external carotid artery

Branch	Main territory
Superior thyroid	Thyroid upper pole, larynx (superior laryngeal), SCM
Ascending pharyngeal	Pharynx, tonsil, middle ear, meninges
Lingual	Tongue, floor of mouth, tonsil (dorsal lingual)
Facial	Face, tonsil (tonsillar branch), submandibular gland, lips
Occipital	Back of scalp, SCM
Posterior auricular	Auricle, scalp, stylomastoid artery to middle ear
Maxillary	Deep face, nose (sphenopalatine), middle meningeal, teeth, middle ear
Superficial temporal	Temporal scalp, parotid

The last four cranial nerves

Nerve	Exit	Motor	Sensory / other	Lesion sign
IX	Jugular foramen	Stylopharyngeus	Posterior third of tongue, tonsil, middle ear, carotid sinus	Lost gag reflex (afferent)
X	Jugular foramen	Pharynx, palate (except tensor), larynx	Larynx, ear (Arnold's), viscera	Uvula deviates to healthy side; cord palsy
XI	Jugular foramen	SCM, trapezius	—	Shoulder drop, weak head turning
XII	Hypoglossal canal	Tongue muscles (except palatoglossus)	—	Tongue deviates to weak side

M Mnemonics

External carotid branches

Some Anatomists Like Freaking Out Poor Medical Students

Superior thyroid, Ascending pharyngeal, Lingual, Facial, Occipital, Posterior auricular, Maxillary, Superficial temporal

Phrenic nerve

C3, 4, 5 keeps the diaphragm alive

Roots of the phrenic nerve

Carotid sheath contents

I See 10 CC's in the IV

Internal jugular vein, vagus (X), Common/internal Carotid

Subclavian first-part branches

VIT

Vertebral, Internal thoracic, Thyrocervical trunk

Thyrocervical trunk

STAIn

Suprascapular, Transverse cervical, Inferior thyroid (which gives the Ascending cervical)

Horner's

PAM

Ptosis, Anhidrosis, Miosis

Q Rapid recall

1. **Level of the carotid bifurcation?** Upper border of the thyroid cartilage (C3–C4).
2. **Branches of the external carotid artery in order?** Superior thyroid, ascending pharyngeal, lingual, facial, occipital, posterior auricular, maxillary, superficial temporal.
3. **How do you tell the ICA from the ECA in the neck?** The ICA has no branches in the neck; the ECA does.
4. **Nerve supply of the carotid sinus and body?** Glossopharyngeal nerve (nerve of Hering).
5. **Relation of the lingual artery and hypoglossal nerve to hyoglossus?** Artery deep to hyoglossus; nerve on its surface.
6. **Roots of the ansa cervicalis?** Superior root C1 (travelling with XII), inferior root C2–C3.
7. **Roots and course of the phrenic nerve in the neck?** C3–C5, on scalenus anterior deep to the prevertebral fascia.
8. **Where does the thoracic duct end?** Junction of the left IJV and subclavian vein.
9. **Branches of the first part of the subclavian artery?** Vertebral, internal thoracic, thyrocervical trunk.
10. **Branches of the thyrocervical trunk?** Inferior thyroid, transverse cervical, suprascapular.
11. **Where does the ECA end?** In the parotid behind the neck of mandible, dividing into maxillary and superficial temporal arteries.
12. **Shamblin classification?** I small and localised with minimal vessel attachment, II partly surrounds, III encases the carotids.
13. **Origin of the IJV?** Continuation of the sigmoid sinus at the jugular foramen.
14. **Which nerve gives the carotid sinus branch?** Glossopharyngeal (IX).
15. **What is Lemierre's syndrome?** IJV thrombophlebitis after pharyngitis, usually Fusobacterium necrophorum, with septic emboli.
16. **Ganglia of the cervical sympathetic chain?** Superior (C2–3), middle (C6), inferior (often fused with T1 as the stellate ganglion).
17. **Signs of Horner's syndrome?** Ptosis, miosis, anhidrosis, apparent enophthalmos.
18. **What is Chassaignac's tubercle?** The anterior tubercle of C6, apex of the scalenovertebral triangle.

Figures from ENT Steps Anatomy (based on Bansal) and Dr. Sameh Doss's Anatomy of Head & Neck handout, for personal study.